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Problem 1

1.1) Notice 1 period $\rightarrow 0.02\,s \rightarrow f = 50\,Hz$.

v_a crosses ϕ at $t = 0.0092\,s$ and is 'sin/cos

Amplitude is 2

$v_a = 2\cdot\cos(\omega t + \varphi) = 2\cdot\cos(2\pi\, 50\,Hz \cdot t + \varphi)$

$v_a\big|_{t=0.0092s} = \phi = 2\cdot\cos(2\pi\, 50\,Hz \cdot 0.0092 + \varphi)$

$\Rightarrow \varphi = \{-255.6^\circ, -75.6^\circ\}$

to choose we evaluate both at $t = \phi$ and one should be -0.5

$\rightarrow \boxed{\varphi = -255.6^\circ}$ $(-75.6^\circ$ is $+0.5)$

$\;=+104.4^\circ$

$v_a = 2\cdot\cos(\omega t + \varphi) = \boxed{2\cos(2\pi 50\cdot t + (-255.6^\circ))}$ $\;=+104.4^\circ$

Notice positive ~~direct~~ sequence (a leads b, b leads c)

$v_b = 2\cdot\cos(2\pi 50 t + (-255.6^\circ) - 120^\circ) = \boxed{2\cos(2\pi 50 t - 15.6^\circ)}$

$v_c = 2\cdot\cos(2\pi 50 t + (-255.6^\circ) - 240^\circ) = \boxed{2\cos(2\pi 50 t - 135.6^\circ)}$

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1.2)

3 phase SV is given by

$\frac{2}{3}\left(f_a + f_b e^{j120} + f_c e^{j240}\right)$

$= \frac{2}{3}\left(2\cos(2\pi 50 t + 104.4^\circ) + 2\cos(2\pi 50 t - \cancel{345.6}\ 15.6^\circ)e^{j120} + 2\cos(2\pi 50 t - 135.6^\circ)e^{j240}\right)$

$= \frac{4}{3}\left(e^{2\pi 50 t + 104.4^\circ} + e^{2\pi 50 t - 15.6^\circ} e^{j120} \ldots\right)$

According to lecture 2 $\quad$ or $2e^{j(2\pi 50 t + 104.4)}$

$\boxed{V_{abc} = 2e^{j(\theta + 104.4^\circ)}} \rightarrow \theta = \omega t \rightarrow$ positive $\rightarrow$ anti-clockwise

1.3)

$v_p = \mathrm{Re}\left(\frac{V_{abc}}{e^{j90^\circ}}\right) = \mathrm{Re}\left(\frac{2e^{j(\theta + 104.4^\circ)}}{e^{j90^\circ}}\right) = 2\,\mathrm{Re}\left(e^{j(\theta + 104.4 - 90)}\right)$

$= \boxed{2\cos(\theta + 14.4)}$ // $2\cos(2\pi 50 t + 14.4^\circ)$

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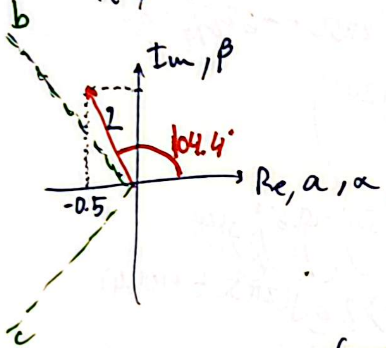
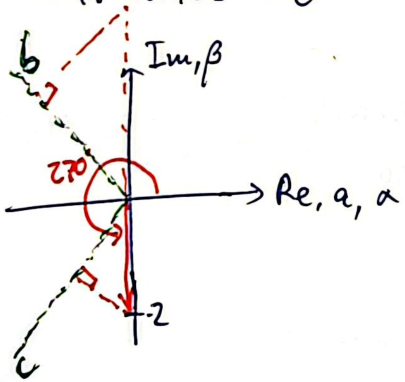
1.4)

$\overline{V_{abc}} = 2e^{j(\theta+104.4)}$

$\overline{V_{abc}}\big|_{t=\phi} = 2\cdot e^{j(\omega t+104.4^\circ)} = 2e^{j(2\pi 50\cdot\phi+104.4^\circ)} = 2e^{j104.4^\circ}$



$\overline{V_{abc}}\big|_{t=0.0092s} = 2e^{j(2\pi 50\cdot 0.0092+104.4)} = 2e^{j(270^\circ)}$



Validated because $a=0, b<0, c>0$ at $t=0.0092$

1.5)

Location of $d = -30^\circ + \theta$ → Rotates at the same speed and direction as the SV.

Location of $q = -30^\circ + 90^\circ + \theta = 60^\circ + \theta$

$$v_d = Re\left(\frac{\overline{V_{abc}}}{e^{j(-30^\circ+\theta)}}\right) = Re\left(\frac{2e^{j(\theta+104.4)}}{e^{j(-30+\theta)}}\right) = -1.4V$$

$$v_q = Re\left(\frac{\overline{V_{abc}}}{e^{j(+60+\theta)}}\right) = +1,4289V$$

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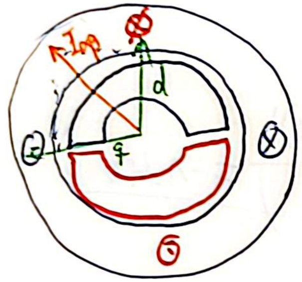
Problem 2

2.1



$i_\alpha = |I_{\alpha\beta}|\cos(\theta - 45^\circ) = \boxed{10\cos(\theta - 45^\circ)}$

$\boxed{i_\beta = 10\sin(\theta - 45^\circ)}$

2.2

$\lambda_\alpha = \lambda_{mpm}\cdot\cos(\theta)$, since at $\theta = \phi$ the magnet and the a coil share flux path

$\lambda_\beta = \lambda_{mpm}\sin(\theta)$

$$T_\alpha = pp\cdot i_\alpha\cdot\frac{d\lambda_\alpha}{d\theta} = 1\cdot 10\cos(\theta-45^\circ)\cdot\frac{d\,\overset{0.12}{\lambda_{mpm}}\cos\theta}{d\theta}$$

$$\boxed{= -1.2\cdot\cos(\theta-45)\cdot\sin(\theta)}$$

$$T_\beta = pp\cdot i_\beta\cdot\frac{d\lambda_\beta}{d\theta} = 1\cdot 10\sin(\theta-45^\circ)\cdot\frac{d\,0.12\cdot\sin(\theta)}{d\theta}$$

$$\boxed{= 1.2\cdot\sin(\theta-45^\circ)\cdot\cos(\theta)}$$

$T_{total} = T_\alpha + T_\beta$

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$T_{total} = -1.2\cos(\theta-45^\circ)\cdot\sin(\theta) + 1.2\sin(\theta-45^\circ)\cos(\theta)$

CAS calculator

$= -1.2\sin(45^\circ) = \boxed{-0.8485\,Nm}$

2.3

$I_q = |I_{\alpha\beta}|\cdot\sin(45), = 5\sqrt{2}\,A$; $I_d = |I_{\alpha\beta}|\cos(45) = 5\sqrt{2}\,A$

$V_q = R\cdot I_q + \overset{SS}{\cancel{\frac{d}{dt} i_q\cdot L_q}} + \omega_r\cdot(L_d\cdot I_d + \lambda_{mpm})$

$V_d = R\cdot I_d + \overset{SS}{\cancel{\frac{d}{dt} i_d L_d}} - \omega_r(L_q I_q)$

$V_q = 1.5\cdot 5\sqrt{2} + 1200\cdot\frac{2\pi}{60}\cdot(5mH\cdot 5\sqrt{2}A + 0.12Wb) = 30.129V$

$V_d = 1.5\cdot 5\sqrt{2} + 1200\,\frac{2\pi}{60}\cdot 5mH\cdot 5\sqrt{2}A = 15.049V$

Peak voltage will be equal to the magnitude of the voltage vector

$\Rightarrow \boxed{V_{\alpha,pk} = \sqrt{V_d^2 + V_q^2} = 33.68V_{pk}}$

2.4) The current expressions remain the same but the flux linkage would be at twice the speed

$\rightarrow i_\alpha = 10\cos(\theta - 45^\circ)$, $i_\beta = 10\sin(\theta - 45^\circ)$

$\lambda_\alpha = \lambda_{mpm}\cos(2\theta)$, $\lambda_\beta = \lambda_{mpm}\sin(2\theta)$

$$T_\alpha = pp \cdot i_\alpha \frac{d\lambda_\alpha}{d\theta} = 2 \cdot 10\cos(\theta - 45) \cdot \frac{d\, 0.06 \cdot \cos(2\theta)}{d\theta}$$

$$T_\alpha = -1.2 \cdot \cos(\theta - 45) \cdot 2 \cdot \sin(2\theta) = -2{,}4\cos(\theta - 45^\circ) \cdot \sin(2\theta)$$

$$T_\beta = 2{,}4 \cdot \sin(\theta - 45) \cdot \cos(2\theta)$$

$$\boxed{T_{total} = T_\alpha + T_\beta = -2.4 \cdot \sin(\theta + 45^\circ)}$$ Not constant! CAS calculator

2.5) The current vector should also rotate at twice the speed

$\rightarrow$ if we consider $i_\alpha = 10\cos(2\theta - 45)$, $i_\beta = 10\cos(2\theta - 45^\circ)$

$$\Rightarrow T_{total} = -2.4\sin(45) = -1{,}697 \text{ Nm}$$

Problem 4

4.1) Since from the datasheet there's no way to calculate slip we will assume $\omega_e = \omega_m \cdot pp$ (negligible slip) at the steady-state operating conditions. This could be due to slip compensation in the control. Anyhow, it would be small (<5%)

~~$\omega_e = \omega_m pp = 2\pi 50 Hz \cdot 2 = 200\pi$ rad/s ≈ 62~~

$$\omega_m = \frac{\omega_e}{pp} = \frac{2\pi 50}{2} = 50\pi = 157 \text{ rad/s}$$

~ 1500 rpm

$$\boxed{T_r = \frac{P_r}{\omega_r} = \frac{2200W}{157 \text{ rad/s}} \approx 14 \text{ Nm}}$$

T_r is probably a bit higher if slip was considered

$\mathrm{acos}(0.8) = 36{,}87^\circ$ inductive

$\rightarrow \bar{I}_s = 4.88 \angle -36.87^\circ$

$\rightarrow \bar{u}_s = R_s \bar{i}_s + \bar{u}_{s\lambda} \rightarrow \bar{u}_{s\lambda} = \bar{u}_s - R_s \bar{I}_s = 220\angle 0^\circ - 3.67 \cdot 4.88\angle -36.87$

$$\Rightarrow \bar{u}_{s\lambda} = 205{,}953\angle +2.99^\circ = i(2\pi 50)\lambda_s$$

$$\boxed{\lambda_s = 0.65557\angle -87{,}009^\circ \text{ Wb RMS} = 0.92711 \text{ Wb}_{pk}}$$

$i_d = 4.88 \cdot Re\left(\frac{e^{j(-36.87^\circ)}}{e^{j2.99}}\right) = \boxed{3,74595A}$

$i_q = 4.88 \cdot Re\left(\frac{e^{j(-36.87^\circ)}}{e^{j(2.99+90)}}\right) = \boxed{-3,12766A}$

4.2)

$\bar{U}_s = R_s \bar{I}_s + \bar{U}_{\lambda s}$

$\bar{U}_{\lambda s} = \bar{U}_s - R_s \bar{I}_s = 173 V_{RMS} - 3.67 \cdot \cancel{4.88}\,3 A \angle -40.4^\circ$

$= 164.77 \angle 2.482^\circ\ Wb_{RMS} = i(2\pi f) \cdot 0.65557$

ξ_s

$\boxed{\Rightarrow f = 40\ Hz}$

4.3)

$\frac{T_{new}}{T_{rated}} = 0.3 = \frac{i_{sd,new}}{i_{sd\,rated}} \Rightarrow i_{sd,new} = 0.3\, i_{sd,rated}$

$= 0.3 \cdot 4.88 A$

$= 1,464 A$

~~Assuming cos φ remains 0.8~~

~~Again, slip is unknown~~

→ Again, slip is neglected

$\omega_m = 900 \cdot \frac{2\pi}{60} = 94,25\ rad/s \rightarrow \omega_e = \omega_m pp = \omega_m \cdot 2 = 188,5\ rad/s$

$\boxed{f_e = \frac{\omega_e}{2\pi} = 30 Hz}$

Problem 5

$P_s = 7500 W$

$\omega_r = 4500\ rpm$

$pp = 2$

$T_L = 6 Nm$

$i_d = 36.7 A$

$i_q = 16 A$

1) The magnitude remains unchanged:

$I_s = \sqrt{36.7^2 + 16^2} = 40.04 A$

~~Non-salient means that without field weakening~~

If it's non-salient, $T = \frac{3}{2} pp \lambda_m i_q$

and λ_m is constant $\rightarrow T \propto i_q$

$\rightarrow i_q^* = 2 \cdot i_q = 32A$

$\rightarrow i_d^* = \sqrt{I_s^2 - i_q^{*2}} = \sqrt{40.04^2 - 32^2} = 24,06 A$

2) $T_{rated} = \frac{P_{rated}}{\omega_r} = \frac{7500W}{4500 \cdot \frac{2\pi}{60}\ rad/s} = 15.9155\ Nm$

$\hookrightarrow \boxed{i_q = \frac{2}{3} \frac{T_{rated}}{pp \cdot \lambda_m} = 42.44A}$

→ The answer depends on the control and the mechanical plant

$T_e - T_{load} = J \frac{d\omega_r}{dt} - b \cdot \omega_r$

acceleration, friction, speed, inertia

For an i_q control, an increase in the load torque would slow down the motor and increase i_q. If $T_e - T_{load} = \emptyset$, the friction should be close to $\emptyset$ to reach the max speed $\frac{V_s}{\lambda_m}$

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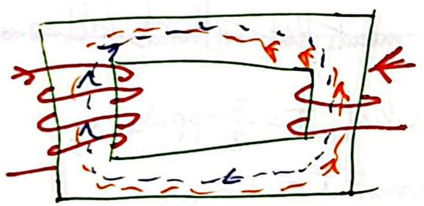
Problem 3

Notice the fluxes produced by both coils are in opposite directions



$\cdots\ \Phi_1$ $\quad$ $---\ \Phi_2$ $\quad$ $N_1 N_2 L_{sgl} = M = M_{12} = M_{21} = \frac{N_1 \phi_{m2}}{i_2} = \frac{N_2 \phi_{m1}}{i_1}$

3.1)

$\lambda_1 = L_{11} \cdot i_1 + M i_2$

$\lambda_2 = L_{22} i_2 + M i_1$

} from slides, same current direction

Our i_2 is in the opposite direction, so let's call it $\boxed{-i_2}$!

In order to satisfy $\frac{N_1 \phi_{m2}}{i_2} = \frac{N_2 \phi_{m1}}{i_1}$, $\boxed{\phi_{m2}}$ must also be negative $\rightarrow$ $\boxed{-\phi_{m2}}$ $=\frac{+}{+}$

$$\boxed{\lambda_1 = L_{11} \cdot i_1 + \frac{N_2 \phi_{m1}}{i_1} \cdot (-i_2) \quad // \quad \lambda_2 = L_{22}(-i_2) + \frac{N_1 \phi_{m2}}{i_2} i_1}$$

~~$\lambda_2 = L_{22} i_2$~~

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3.2)

$i_1' = \frac{N_1}{N_2} i_1 = 2 \cdot i_1$

$L_m = N_2^2 L_{sgl} = L_{22} \Rightarrow L_{sgl} = \frac{L_{22}}{N_2^2}$

$L_{11} = \left(\frac{N_1}{N_2}\right)^2 L_m = 4 L_{22}$

$\lambda_1' = 4 L_{22} \cdot 2 \cdot i_1 - M \cdot i_2 = 8 L_{22} \cdot i_1 - N_1 N_2 \cdot L_{sgl} \cdot i_2$

$\lambda_1' = 8 L_{22} i_1 - N_1 N_2 \frac{L_{22}}{N_2^2} \cdot i_2 = 8 L_{22} i_1 - \frac{N_1}{N_2} \cdot L_{22} \cdot i_2$

$$\boxed{= L_{22} \cdot \left(8 i_1 - \frac{N_1}{N_2} i_2\right) = \lambda_1'}$$